

Benchmarking Exported Key Material from Commercial QKD Systems Using SENTRY-Q: A Model-Based Output Validator

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Abstract—Quantum Key Distribution (QKD) enables two parties to establish fresh cryptographic key material with information-theoretic security guarantees, given an authenticated classical channel and appropriate device and threat models. As QKD deployments mature from laboratory settings into production-grade field infrastructure, a practical gap emerges: protocol-level metrics such as quantum bit error rate (QBER) and secret key rate (SKR) characterise the quantum link but do not directly specify how exported key blocks as consumed by downstream key management systems (KMS) and cryptographic applications should be validated for stable, anomaly-free behaviour at the delivery interface. This paper addresses that operational gap. We present an anonymised benchmark study of three commercial QKD systems using SENTRY-Q, a reproducible measurement workflow that computes five block-level indicators Hamming weight balance, min-entropy proxy, Lempel–Ziv complexity, Borel normality deviation, and serial correlation complemented by a long-stream NIST SP 800-22 sanity check applied to the concatenated key pool. The study covers $N=10,000$ exported 256-bit keys per system, spanning a laboratory DV-QKD link (System-1, ~ 20 km), a dark-fibre field DV-QKD deployment (System-2, ~ 100 km), and a laboratory CV-QKD system (System-3). We scope the contribution as model-based output benchmarking, not as a proof of conditional secrecy. Within that scope, all three systems exhibit block-level distributions consistent with unbiased reference expectations with Hamming weight medians of exactly 128.0 bits and min-entropy medians of 241.85 bits across all systems however, the NIST long-stream sanity check reveals system-differentiated anomalies: a Non-Overlapping Template Matching failure in System-2 and an Overlapping Template Matching failure with borderline Binary Matrix Rank in System-3, that are invisible to block-level analysis. These anomalies do not constitute proven security vulnerabilities; rather, they represent operationally significant signals that warrant engineering investigation. Critically, the block-level and long-stream analysis layers detect fundamentally different failure classes and cannot substitute for one another, both are necessary components of a complete key-delivery-pipeline benchmarking workflow. We discuss deployment implications and provide a standardised regression-testing artefact suitable for acceptance and longitudinal monitoring workflows.

Index Terms—QKD, key-material benchmarking, SENTRY-Q, NIST SP 800-22, Lempel–Ziv complexity, min-entropy, privacy amplification, regression testing, operational assurance.

I. INTRODUCTION

Quantum Key Distribution (QKD) allows two parties to establish fresh shared key material with security rooted in quantum mechanics rather than computational hardness assumptions [1]. Protocol-level performance is routinely characterised by the quantum bit error rate (QBER), secret key rate (SKR), and,

where available, finite-key security parameters [2], [3]. As QKD transitions from controlled demonstrations into operational metropolitan and long-haul field deployments [4], [5], these metrics remain necessary but become insufficient on their own: a distinct and underappreciated failure class arising in the software, firmware, and integration logic of the key delivery pipeline rather than in the quantum layer itself can degrade exported key material even when the quantum link is stable and the underlying protocol executes correctly.

Software and integration faults well-documented in classical cryptographic engineering including block truncation, unintended padding, key duplication or replay after system restarts, block mis-ordering from buffer management, and mismatches between the privacy amplification output format and the KMS input format [6], [7] all operate at the export interface. They are therefore outside the scope of protocol-level security proofs, invisible to link-layer telemetry, and potentially persistent across sessions unless specifically monitored. As QKD deployments scale across heterogeneous nodes and vendors, the probability of encountering such faults grows.

This paper addresses the following practical question: *What does well-behaved exported QKD key material look like at the block level, and how can deployments benchmark it reproducibly?* We present an anonymised benchmark study of three commercial QKD systems, a laboratory DV-QKD link at ~ 20 km (System-1), a dark-fibre field DV-QKD deployment at ~ 100 km (System-2), and a laboratory CV-QKD system (System-3), using SENTRY-Q [8], a web-based measurement workflow. We evaluate pools of $N=10,000$ exported 256-bit keys per system using five block-level indicators and a concatenated long-stream NIST sanity check. As we demonstrate, these two analysis layers detect fundamentally different failure classes: the long-stream layer reveals system-differentiated anomalies that are entirely invisible to all five block-level indicators, directly motivating the two-layer evaluation architecture of SENTRY-Q.

We state the scope precisely: output-level statistical testing of privacy-amplified key material cannot certify conditional secrecy against an adversary and cannot validate protocol-level security parameters. Strong post-processing can render outputs statistically indistinguishable from ideal randomness even when upstream assumptions are violated. The objective of this work is therefore *not* to prove security from output bits, but to

provide a reproducible, model-based benchmarking methodology capable of detecting gross delivery-path anomalies, characterising pipeline stability, and supporting acceptance and regression testing a complementary, not competing, contribution to protocol-level security accounting.

The paper makes four concrete contributions:

- 1) An anonymised, comparative benchmark of exported 256-bit key material from three commercial QKD systems spanning two QKD paradigms (DV and CV), two deployment environments (laboratory and field), and a five-fold range in link distance. The benchmark reveals system-differentiated anomalies in the NIST long-stream sanity check, a Non-Overlapping Template Matching failure in System-2 and an Overlapping Template Matching failure with borderline Binary Matrix Rank in System-3, that are absent from all block-level indicators across all three systems.
- 2) A model-based, finite-sample evaluation methodology for short key blocks, providing reference-expectation baselines and explicit operational criteria for distinguishing expected statistical extremes from anomalies warranting investigation.
- 3) A reproducible measurement workflow (SENTRY-Q) generating standardised, machine-parseable benchmarking artefacts suitable for regression testing across software/hardware revisions and for longitudinal operational monitoring.
- 4) A demonstration that block-level distributional indicators and long-stream concatenated-stream sanity checks detect fundamentally different failure classes and cannot substitute for one another, establishing that both layers are necessary components of a complete key-delivery-pipeline benchmarking workflow.

II. BACKGROUND AND RELATED WORK

A. Protocol-Level Security Frameworks

QKD security is formalised in composable frameworks that bound an adversary’s accessible information as a function of observable channel parameters [9], [10]. Finite-key analyses [11], [12] extend these frameworks to practical regimes where limited sample sizes constrain achievable key lengths. These frameworks correctly address the question of conditional secrecy but do not directly specify whether exported key material remains stable after traversing the post-processing and delivery pipeline, the question targeted by this work.

B. Field Deployments and Operational Networks

Extensive literature documents metropolitan and multi-node QKD deployments including the Tokyo QKD Network [5], the SECOQC trusted-repeater network [13], and the SwissQuantum long-duration field trial [4] primarily reporting link-level metrics (QBER, SKR, uptime) and network functionality. Standardised characterisation of exported short key blocks, with explicit finite-sample interpretation and delivery-path anomaly detection, receives comparatively limited treatment in this body of work.

C. Key Delivery Interfaces and Key Management

ISO/IEC 23837-1:2023 [14] and its companion evaluation methods document ISO/IEC 23837-2:2023 [15] represent the current international standard for QKD security requirements and test methods. While comprehensive in their treatment of physical layer security, protocol correctness, and key delivery interface functionality, neither document specifies requirements or methods for statistical characterisation of exported key material at the delivery interface. The export boundary is treated as a functional trust point rather than a statistical test point, with correctness implicitly assumed to follow from protocol-level compliance. SENTRY-Q directly addresses this gap, providing the output-level statistical benchmarking methodology that the current standards landscape does not cover. ETSI GS QKD 004 and QKD 014 [16], [17] define application-facing key delivery APIs and treat the export interface as the boundary between a QKD system and consuming applications. Recent work on QKD-KMS integration [18] highlights the complexity and potential fragility of this interface, motivating regression-style validation at the block level as proposed here.

D. Statistical Testing and Entropy Assessment

NIST SP 800-22 [19] and NIST SP 800-90B [20] provide widely used guidance for randomness testing and entropy-source assessment respectively. Both standards are designed primarily for long sequences or entropy sources rather than short fixed-length delivered blocks. Our methodology explicitly accounts for this by reserving NIST SP 800-22 for the concatenated pool stream and using purpose-designed block-level indicators with finite-sample reference models for individual key evaluation. Specifically, NIST STS is *not* applied to individual 256-bit blocks, it is applied only to the concatenated 2.56 Mbit stream and reported as a single-stream sanity check, consistent with NIST’s minimum recommended sequence length.

E. Implementation Evaluation and the Gap Addressed

The practical QKD security community increasingly focuses on implementation-level certification and attack-surface evaluation [21], [22]. These efforts target exploitable hardware loopholes and are complementary to our contribution. The practical gap we address, standardised, model-based benchmarking of exported short key blocks with finite-sample interpretation and repeatable regression artefacts, has received comparatively little attention despite its direct operational relevance as QKD deployments scale.

III. SCOPE, NON-GOALS, AND LIMITATIONS

A. Evaluation Object and Benchmarking Perspective

We evaluate application-facing key material at the export interface the 256-bit blocks delivered to a downstream KMS or application treating this material as the observable artefact available to operators. We apply model-based testing: observed distributions are interpreted relative to reference expectations

derived from idealised models to (i) quantify typical finite-sample variability and (ii) identify deviations unlikely under baseline assumptions. Table I states the precise scope boundary.

TABLE I: Scope boundary of this benchmark study.

This paper ESTABLISHES	This paper does NOT establish
✓ Reproducible, model-based benchmarking of exported QKD key blocks	× Proof of conditional secrecy or information-theoretic security
✓ Detection of gross delivery-path anomalies (truncation, padding, duplication, replay)	× Validation of protocol-level security parameters (QBER bounds, finite-key lengths)
✓ Finite-sample reference-model interpretation of rare block-level extremes	× Root-cause attribution to quantum layer, post-processing, or interface
✓ Regression baseline artefacts for acceptance and longitudinal monitoring	× Full NIST multi-sequence pass-proportion evaluation
✓ Cross-paradigm comparison (DV-QKD vs. CV-QKD)	× Certification against side-channel or injection attacks

B. What This Analysis Cannot Establish

Output-level benchmarking cannot distinguish a correctly operating system from one where upstream assumptions are violated but privacy amplification produces statistically indistinguishable output. It cannot attribute anomalies to specific hardware or channel components without additional system telemetry, and does not constitute a full entropy-source assessment under NIST SP 800-90B.

C. Anonymisation Policy

All systems are reported with anonymised identifiers. Product names, firmware versions, and vendor-specific configuration parameters enabling attribution are withheld to ensure vendor-neutral comparison.

IV. MODEL-BASED BENCHMARKING METHODOLOGY

A. Notation and Data Organisation

For each system, we analyse a pool of $N=10,000$ exported keys $K^{(j)} \in \{0,1\}^{256}$, $j = 1, \dots, N$, with block length $L=256$ bits. All five primary indicators are computed per-block independently and aggregated across the pool. For long-stream analysis we form the concatenated bitstream $S = K^{(1)} \| K^{(2)} \| \dots \| K^{(N)}$ of nominal length $n = N \cdot L = 2,560,000$ bits. The effective concatenated length n_{eff} equals the number of bits remaining after integrity validation, any shortfall is logged and reported.

B. Reference Expectations and Finite-Sample Interpretation

Because $L=256$ is small, finite-sample variability at the block level is substantial. A core methodological commitment is that all block-level observations are interpreted relative to reference expectations from idealised models. We use two complementary operational criteria:

- **Per-block tail criterion:** block $K^{(j)}$ is flagged if its indicator's two-sided tail probability under the reference model falls below a chosen level α .
- **Batch-frequency criterion (primary):** an event is operationally extreme if its expected count in $N=10,000$ blocks satisfies $N \cdot \Pr(\cdot) < 1$ under the reference model. This directly answers: "Should we expect to see this in a healthy pool of 10,000 blocks?" It is the primary criterion used throughout this paper. The per-block tail criterion alone generates false alarms at a rate fully consistent with healthy systems, at $\alpha=0.001$, approximately 10 blocks per pool are expected to be flagged by chance.

C. Block-Level Indicators

For each block $K^{(j)}$ we compute five indicators, reported as distributions over the pool with summary statistics (median, quantiles, extremes). Table II summarises all indicators with their reference models and known limitations.

1) *Hamming Weight Distribution:* Let $X^{(j)} = \sum_i K_i^{(j)}$ count the set bits in block j . Under the i.i.d. reference model, $X \sim \text{Bin}(256, \frac{1}{2})$ with mean 128 and variance 64. We compare the empirical distribution $\{X^{(j)}\}$ to this reference and interpret extremes via the batch-frequency criterion. The standard error of the empirical mean over $N=10,000$ blocks is $\sigma/\sqrt{N} = 8/100 = 0.08$ bits, giving adequate resolution to detect systematic mean deviations of a fraction of one bit.

2) *Min-Entropy Indicator:* For each block $K^{(j)}$, SENTRY-Q computes an upper bound $\hat{p}_{\max}^{(j)}$ on the probability of the most predictable outcome under a fixed estimator configuration, yielding the per-block score

$$\hat{H}_{\min}^{(j)} = -L \log_2(\hat{p}_{\max}^{(j)}). \quad (1)$$

The same estimator configuration is used for all three systems. $\hat{H}_{\min}$ is an operational benchmarking proxy, not a proof-level secrecy certificate, its pool distribution serves as a regression baseline.

3) *Lempel–Ziv Complexity:* We compute the normalised LZ76 score [23]

$$C_{\text{LZ}}^{(j)} = \frac{c(m) \log_2 m}{m}, \quad (2)$$

where $c(m)$ is the number of distinct phrases under incremental LZ76 parsing of the 256-bit block. High scores indicate the absence of repetitive structure or compressibility artefacts.

4) *Borel Normality:* For each block we compute the maximum overlapping k -gram deviation

$$\Delta_k^{(j)} = \max_{u \in \{0,1\}^k} |\hat{f}^{(j)}(u) - 2^{-k}|, \quad k \in \{1, 2\}, \quad (3)$$

where $\hat{f}^{(j)}(u)$ is the empirical frequency of pattern u over $L-k+1$ overlapping k -grams. Low pooled Δ_k indicates consistent k -gram frequency uniformity.

TABLE II: Summary of block-level indicators and long-stream sanity tests.

Indicator	Level	Input	Purpose	Reference	Output	Known Limitation
Hamming weight	Block	$10^4 \times 256$ b	Bit-balance check, tail interpretation	$\text{Bin}(256, \frac{1}{2})$	Histogram + quantiles + tail-freq.	Marginal bit bias only, no higher-order structure
Min-entropy $\hat{H}_{\min}$	Block	$10^4 \times 256$ b	Worst-case predictability proxy	Block-length upper bound	$\hat{H}_{\min}$ distribution + stats	Estimator upper bound, NOT a secrecy certificate
Lempel-Ziv C_{LZ}	Block	$10^4 \times 256$ b	Repetitive structure / compressibility	Max C_{LZ} for random 256-b strings	C_{LZ} distribution + stats	Short-block LZ76, limited sensitivity near max
Borel normality Δ_k ($k=1, 2$)	Block	$10^4 \times 256$ b	1- and 2-gram frequency deviation	Uniform 2^{-k} expectation	Δ_1, Δ_2 distributions + stats	Overlapping k -grams, finite-block resolution limited
Serial correlation ρ_1	Block	$10^4 \times 256$ b	Lag-1 within-block bit dependence	$\mathbb{E}[\rho_1] \approx 0$ for i.i.d. bits	ρ_1 distribution + stats	Lag-1 only, higher-order dependencies not captured
NIST STS (concatenated)	Stream	n_{eff} bits	Global sanity check on pooled bitstream	Uniform p -values under H_0	Per-test p -values vs. $\alpha=0.01$	Single-stream mode, not full multi-sequence NIST methodology

5) *Serial Correlation Coefficient*: We compute the within-block lag-1 Pearson correlation

$$\rho_1^{(j)} = \frac{\sum_{t=1}^{L-1} (K_t^{(j)} - \bar{K}^{(j)}) (K_{t+1}^{(j)} - \bar{K}^{(j)})}{\sum_{t=1}^L (K_t^{(j)} - \bar{K}^{(j)})^2}, \quad (4)$$

where $\bar{K}^{(j)}$ is the mean bit value of block j . Under the i.i.d. model, $\mathbb{E}[\rho_1] \approx 0$ for large L . Note that ρ_1 captures only lag-1 dependence, higher-order bit dependencies are not detectable by this indicator alone.

D. Long-Stream Sanity Tests

Block-level indicators are designed for delivery-path anomaly localisation at individual key granularity. To detect structure emerging only at the pool scale, we apply two complementary global sanity checks to the concatenated stream S :

- **NIST SP 800-22 (single-stream summary)**: all 15 NIST STS tests applied to S with $\alpha=0.01$. We do not claim compliance with the full multi-sequence pass-proportion methodology.

V. SENTRY-Q: MEASUREMENT WORKFLOW

A. Inputs, Validation, and Configuration

SENTRY-Q accepts a pool of exported keys as fixed-length binary strings. For this study, each run targets $N=10,000$ keys of $L=256$ bits. The workflow applies uniform basic integrity validation (expected block length, binary encoding, absence of all-zero or all-one blocks indicative of padding or initialisation artefacts). Blocks failing validation are excluded from concatenated-stream tests, n_{eff} is logged per run. All five block-level indicators use an identical, parameterless configuration across all three systems.

B. Block-Level Analysis Pipeline

For each validated block $K^{(j)}$, SENTRY-Q computes the five indicators sequentially. Computed per-block values are streamed to a pooled aggregator maintaining running statistics (median, 25th/75th/90th percentiles, min, max) without retaining individual block data beyond the computation window.

C. Concatenated-Stream Analysis and Reporting Artefacts

SENTRY-Q concatenates all validated blocks into S and invokes NIST STS as external processes with standardised inputs. A standard run produces: (i) block-level distribution plots with reference overlays, (ii) per-indicator pooled summary tables, (iii) concatenated-stream NIST STS, and (iv) a machine-parseable, version-stamped artefact bundle enabling regression comparison across software/hardware revisions.

VI. EXPERIMENTAL SETUP AND DATASETS

A. Systems Under Test

We benchmark three fibre-based QKD systems operated under steady-state vendor-recommended conditions, as detailed in Table III. The diversity in QKD paradigm, deployment environment, and link distance is deliberate: it enables assessment of whether SENTRY-Q produces interpretable and comparable results across heterogeneous systems.

TABLE III: Anonymised systems under test.

System	Type	Protocol	Environment	Link
System-1	DV-QKD	BBM92	Lab (controlled)	~20 km
System-2	DV-QKD	BBM92	Dark-fibre field	~100 km
System-3	CV-QKD	BB84	Lab (controlled)	Short fibre

System-1 is a well-controlled laboratory baseline providing a low-noise reference. System-2 introduces additional complexity vibration, temperature cycling, splice losses at ~100 km field

distance. System-3 introduces a qualitatively distinct QKD paradigm (coherent-detection CV-QKD) with a different noise model and post-processing chain.

B. Data Collection Protocol

For each system, exported key material is collected during steady-state operation and recorded in delivery order without reordering, filtering, or selection. All data is ingested by SENTRY-Q with no transformation beyond binary format normalisation. Identical SENTRY-Q configuration is applied to all three systems.

C. Statistical Power

For block-level indicators, $N=10,000$ gives a standard error of $\sigma/\sqrt{N} = 0.08$ bits on the Hamming weight mean, providing adequate power to detect systematic biases of a fraction of one bit. For NIST STS, $n_{\text{eff}} = N \cdot L = 2,560,000$ bits meets or exceeds NIST's recommended minimum sequence length for all 15 tests [19]. We acknowledge $N=10,000$ is a single temporal snapshot, multi-session longitudinal evaluation is a direction for future work.

VII. RESULTS

A. Hamming Weight Distribution

The empirical Hamming weight histograms for all three systems closely match the $\text{Bin}(256, \frac{1}{2})$ reference curve. Key observations:

- Empirical medians are exactly 128.0 in all three systems, matching the theoretical expectation $L/2 = 128$.
- Observed ranges (System-1: 98–156, System-2: 93–161, System-3: 97–164) are consistent with finite-sample expectations. Under the batch-frequency criterion, values at or below ≈ 93 or above ≈ 163 are expected fewer than once per $N=10,000$ blocks, the single extreme blocks observed in Systems 2 and 3 are within expected statistical variation.
- No systematic skew, bimodality, or truncation artefacts are visible, confirming the absence of bit-balance delivery-path faults.

B. Block-Level CDF Indicators

Table IV presents the pooled summary statistics.

TABLE IV: Pooled block-level indicator statistics ($N=10,000$ keys, $L=256$ bits per system).

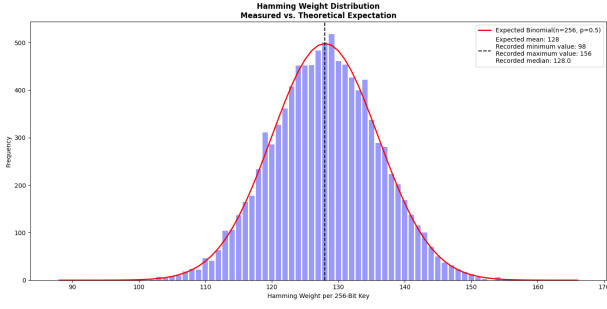
Indicator	Sys-1	Sys-2	Sys-3
HW median (bits)	128.0	128.0	128.0
HW range	98–156	93–161	97–164
$\hat{H}_{\min}$ mean (bits)	238.1	238.4	238.4
$\hat{H}_{\min}$ median (bits)	241.8	241.8	241.8
$\hat{H}_{\min}$ std dev	13.07	13.04	12.89
C_{LZ} mean	58.17	58.19	58.18
C_{LZ} median	58	58	58
C_{LZ} std dev	0.74	0.73	0.74
Δ_1 mean	0.037	0.037	0.037
Δ_1 median	0.034	0.034	0.034
Δ_1 std dev	0.021	0.020	0.020
ρ_1 mean	0.050	0.049	0.050
ρ_1 median	0.043	0.043	0.043
ρ_1 std dev	0.037	0.036	0.037

1) *Min-Entropy*: The min-entropy CDF profiles are nearly identical across all three systems: mean $\hat{H}_{\min}$ of 238.14, 238.36, and 238.37 bits respectively, with a common median of 241.85 bits and standard deviations of 12.89–13.07 bits. The ~ 18 -bit deficit relative to the block length of 256 bits is attributable to finite-sample estimation bias in the per-block estimator [24] and is consistent with expected values for genuinely random 256-bit blocks under the applied estimator configuration.

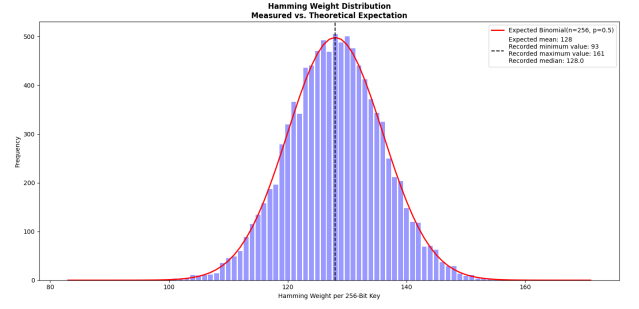
2) *Lempel–Ziv Complexity*: LZ complexity scores are strikingly consistent: means of 58.17, 58.19, and 58.18 with identical standard deviations of 0.74 across all systems. The bulk of keys score 57, 58, or 59, indicating near-maximum incompressibility and confirming the absence of repetitive sub-block structure.

3) *Borel Normality*: Borel normality CDFs exhibit a right-skewed distribution with medians of 0.034 and means of ≈ 0.037 across all three systems, consistent with the theoretical distribution of maximum k-gram deviation for genuinely random bit sequences. No systematic inter-system differences are observed.

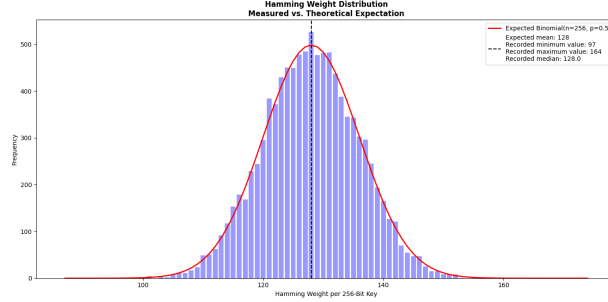
4) *Serial Correlation*: Serial correlation CDFs show broad, right-skewed distributions spanning 0 to ≈ 0.230 – 0.253 across systems, consistent with the absence of systematic lag-1 bit dependencies. System-2 shows a marginally lower mean (0.0492 vs. ≈ 0.0503 – 0.0507) within expected statistical variation.



(a) System 1



(b) System 2



(c) System 3

Fig. 1: Measured versus theoretical Hamming weight distributions for the three QKD systems.

C. NIST SP 800-22 Long-Stream Sanity Check

Table V presents the full NIST STS results for all three systems on the concatenated $n_{\text{eff}} \approx 2,560,000$ -bit streams. These results are global sanity checks on the pooled bitstream, they are not per-key security certificates.

1) *System-1: Full Pass*: System-1 passes all 15 NIST STS tests with p -values comfortably above $\alpha=0.01$. System-1 provides the benchmark reference.

2) *System-2: Non-Overlapping Template Matching Failure*: System-2 passes 14 of 15 NIST STS tests but fails Non-Overlapping Template Matching ($p < 0.01$). This indicates that certain short bit patterns are systematically over- or under-represented in the System-2 concatenated key bitstream. Critically, this anomaly is entirely invisible to all five block-level indicators. Systems 1 and 2 produce statistically identical block-level distributions. The anomaly emerges only at the 2.56 Mbit concatenated-stream scale, directly motivating the two-layer evaluation architecture of SENTRY-Q. A single test failure at $\alpha=0.01$ carries a 1% false-positive probability, however, System-2 fails in a specific test dimension that System-1 passes under identical conditions, which substantially reduces the probability of a Type I error.

3) *System-3: Overlapping Template Matching Failure*: System-3 fails Overlapping Template Matching (p effectively zero) and produces a borderline Binary Matrix Rank result ($p=0.068$, above threshold but flagged for regression monitor-

TABLE V: NIST SP 800-22 single-stream results ($n_{\text{eff}}=2,560,000$ bits, $\alpha=0.01$). **Green** = pass, **red** = fail, **amber** = borderline ($*p=0.068$).

NIST Test	Sys-1	Sys-2	Sys-3
Monobit	0.570	0.570	0.643
Frequency Within Block	0.420	0.420	0.434
Runs	0.516	0.516	0.529
Longest Run Ones In Block	0.107	0.107	0.153
Binary Matrix Rank	0.622	0.622	0.068*
Discrete Fourier Transform	0.731	0.731	0.119
Non-Overlapping Template	0.547	< 0.01	0.605
Overlapping Template	0.935	0.935	< 0.01
Maurer's Universal	0.748	0.748	0.987
Linear Complexity	0.522	0.522	0.790
Serial	0.495	0.495	0.182
Approximate Entropy	0.575	0.575	0.946
Cumulative Sums	0.541	0.541	0.424
Random Excursion	0.422	0.422	0.578
Random Excursion Variant	0.448	0.448	0.667
Overall	Pass	1 fail	1 fail + 1 border-line

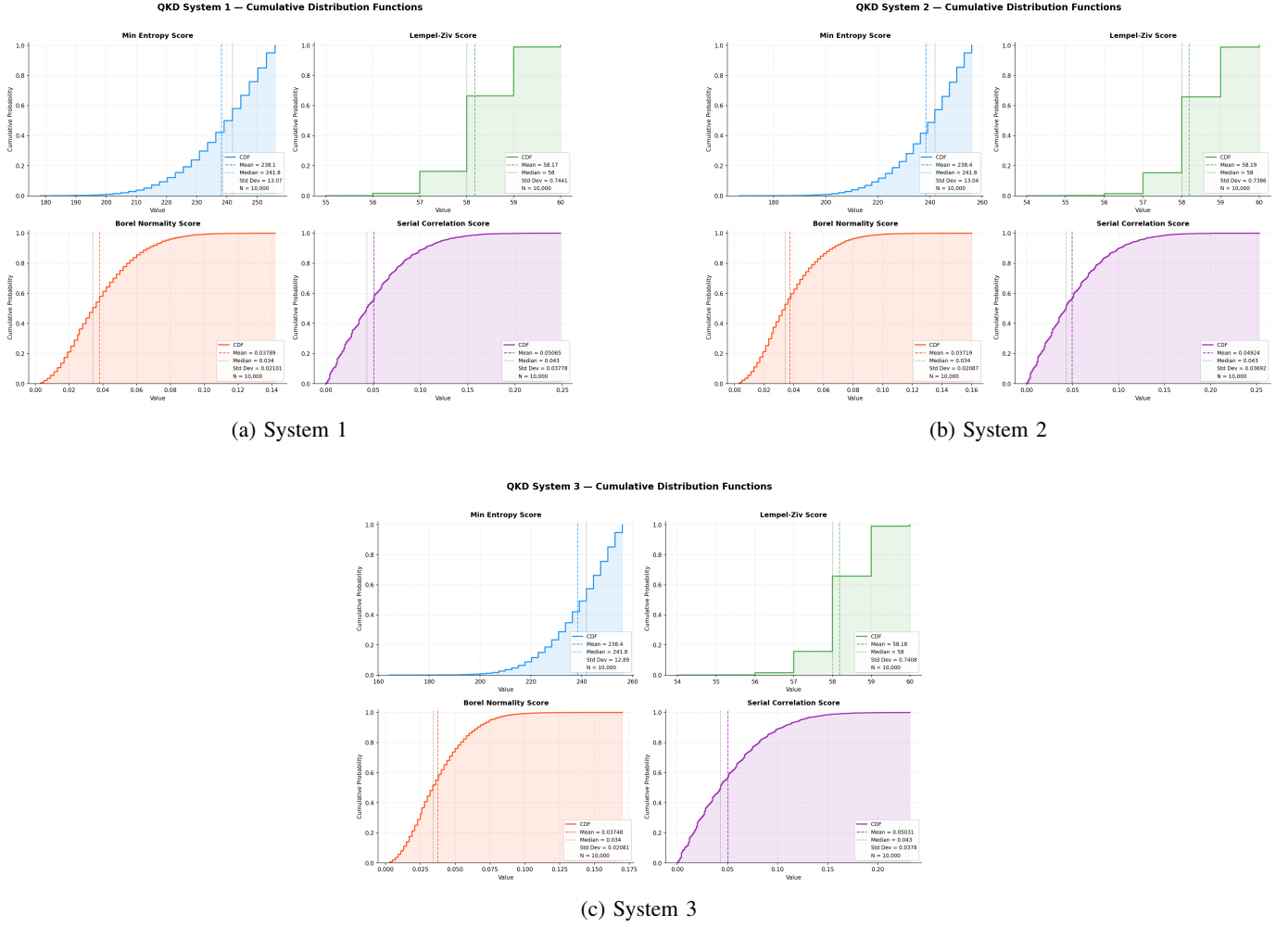


Fig. 2: CDF distributions for the three evaluated systems.

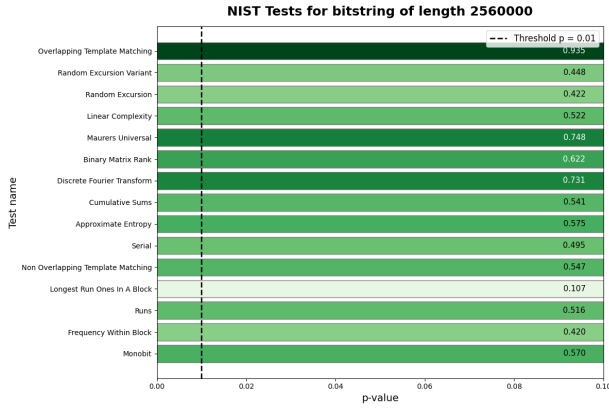
ing). The Overlapping Template Matching failure is sensitive to periodic or near-periodic bitstream structure. The borderline Binary Matrix Rank suggests possible linear dependencies in the key bit matrix, a property consistent with structured linear maps such as Toeplitz matrices used in some implementations of the privacy amplification hash function [25]. As with System-2, these anomalies are absent from all block-level CDF indicators.

VIII. DISCUSSION

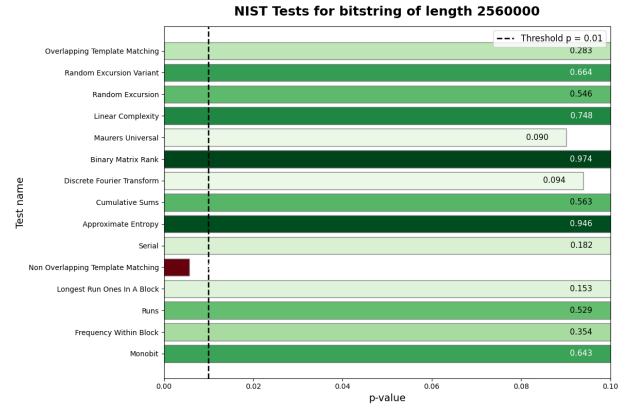
A. Concordance of Block-Level Indicators

The near-perfect concordance of all five block-level indicators across all three systems identical Hamming weight medians, common min-entropy median of 241.85 bits, and identical LZ standard deviations of 0.74 confirms that privacy amplification is performing its core randomisation and entropy extraction function consistently in all three implementations.

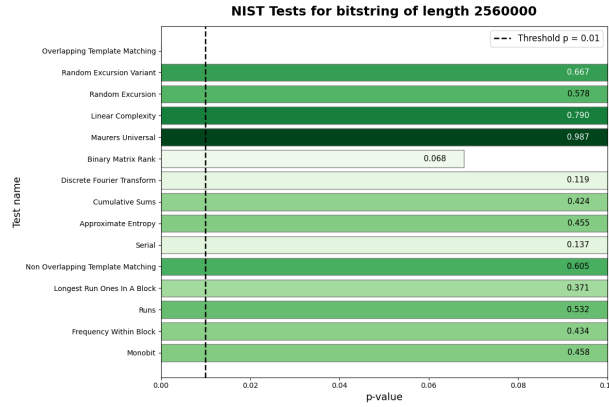
This concordance also demonstrates a fundamental limitation: block-level distributional metrics are blind to pattern-level structure that emerges at the pool or bitstream scale. The NIST failures in Systems 2 and 3, invisible to all five block-level indicators, illustrate that individually unremarkable blocks can collectively exhibit detectable structure in their concatenation. The two layers of analysis detect fundamentally different failure classes and are both necessary components of a complete benchmarking workflow.



(a) System 1



(b) System 2



(c) System 3

Fig. 3: NIST Tests for the three evaluated systems.

B. Delivery-Path Hypotheses

Within the scope of output-level benchmarking, we identify plausible delivery-path hypotheses for the observed anomalies. For System-2, the Non-Overlapping Template Matching failure is consistent with: (a) a bias in the universal hash family used for privacy amplification, (b) residual raw-key structure not fully eliminated at the current compression ratio, or (c) an implementation fault in the output serialisation or buffering layer. For System-3, the Overlapping Template Matching failure and borderline Binary Matrix Rank are consistent with: (a) an insufficiently random or reused Toeplitz matrix seed in the privacy amplification function, (b) an improperly seeded hash function, or (c) a periodic artefact in the CV-QKD post-processing chain. Distinguishing these hypotheses requires system telemetry beyond the export interface. We emphasise that these anomalies represent detectable departures from the unbiased reference model that a correctly functioning delivery pipeline should not exhibit; whether they constitute security-relevant weaknesses requires additional analysis outside the scope of this work.

C. Implications for Operational Deployment

- 1) **Adopt a two-layer validation architecture.** Block-level indicators provide anomaly localisation at key-block granularity. Long-stream NIST sanity checks provide global structure detection. The two layers detect different failure classes and cannot substitute for each other.
- 2) **Establish SENTRY-Q baselines before production deployment.** Operators of Systems 2 and 3 should investigate the flagged anomalies, remediate, and re-run to establish a clean post-fix baseline.
- 3) **Account for paradigm-specific behaviour in CV-QKD.** System-3 exhibits a qualitatively different NIST failure signature from the DV systems. Benchmarking workflows should be calibrated against CV-QKD-specific reference models.

IX. CONCLUSION

We have presented an anonymised, model-based benchmark study of exported 256-bit key material from three commercial QKD systems using SENTRY-Q. Across all three systems, the five block-level indicators confirm that privacy amplification is performing its core function correctly: Hamming weight medians are exactly 128.0, min-entropy medians are 241.85 bits,

and Lempel–Ziv complexity scores approach the theoretical maximum. These results confirm delivery-path bit-balance and structural integrity at the block scale.

The NIST SP 800-22 long-stream sanity check reveals system-differentiated anomalies invisible to block-level analysis: System-2 fails Non-Overlapping Template Matching, and System-3 fails Overlapping Template Matching with a borderline Binary Matrix Rank result. These anomalies, emerging only at the 2.56 Mbit concatenated-stream scale, are the central empirical finding of the paper: block-level distributional metrics and long-stream sanity checks detect fundamentally different failure classes, and both layers are necessary for complete key-delivery-pipeline benchmarking.

Output-level benchmarking is not a substitute for protocol-level security accounting, and the detected anomalies do not constitute proven security vulnerabilities; rather, they constitute operationally significant signals warranting engineering investigation. SENTRY-Q provides the reproducible, machine-parseable benchmarking artefact needed to support such investigations and to establish regression baselines for acceptance and longitudinal monitoring in production QKD deployments.

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